Manned missions

In the course of six decades, the idea of human spaceflight has gone from the unthinkable to the almost routine (though still inevitably risky). So far more than 550 people have flown into space, the vast majority of them aboard either the United States' Space Shuttle or Russia's Soyuz spacecraft. In future decades, tourism may see at least brief trips into space become commonplace.

WOW!

In 1961, when President Kennedy committed the US to reaching the moon by the end of the decade, NASA had just 15 minutes of human spaceflight experience.

The first man in space

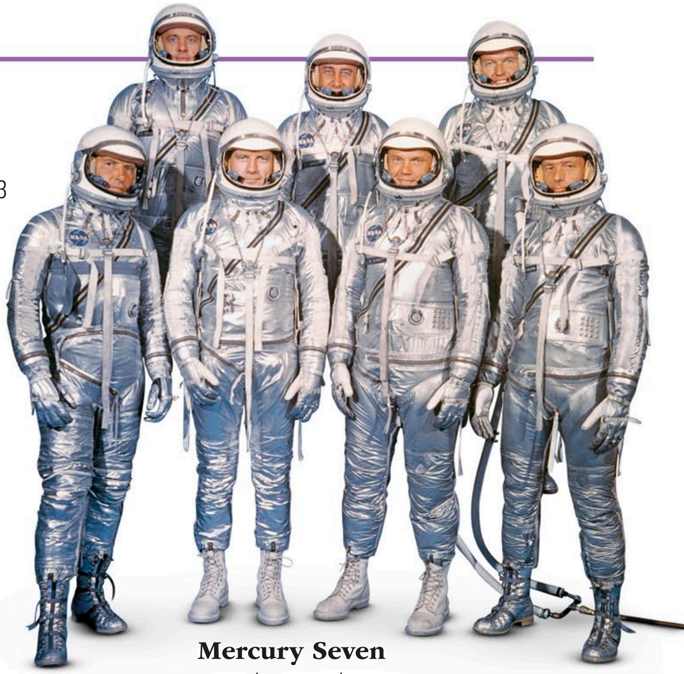
- **What?** Vostok 1
- **Who?** Yuri Gagarin
- **Where and when?** Russia, 1961

Until the 1950s, no one knew what effect spaceflight might have on a human being. Before risking a cosmonaut, Russia had to prove that it was possible to survive by launching several spacecraft with dogs on board. Yuri Gagarin's first record-breaking flight was limited to a single orbit of Earth lasting 108 minutes. In 1963, the first woman in space, Valentina Tereshkova, orbited Earth 48 times for more than 70 hours in Vostok 6.

Americans into orbit

- **What?** Mercury program
- **Who?** Seven US astronauts (six of whom flew on Mercury missions)
- **Where and when?** US, 1958–1963

Hampered by less powerful rockets, NASA's Mercury program lagged behind the Russians. The first American in space, Alan Shepard, could only make a "hop" into space that failed to reach orbit in May 1961. In February 1962, the Atlas rocket was ready to take John Glenn into orbit aboard the Mercury capsule Friendship 7.



Mercury Seven astronauts

A Soyuz capsule in 2015, docked to the International Space Station

Soyuz modules

- **What?** Soyuz spacecraft
- **Who?** OKB-1 design bureau/Energia Corp.
- **Where and when?** Russia, 1967–present

First launched in 1967 (on a disastrous mission that killed its pilot), the Soyuz spacecraft was nevertheless a major advance. It was the first with three modules, including specialized capsules for working in orbit and returning to Earth. Thanks to several generations of updates, it has been the backbone of the Russian space program.